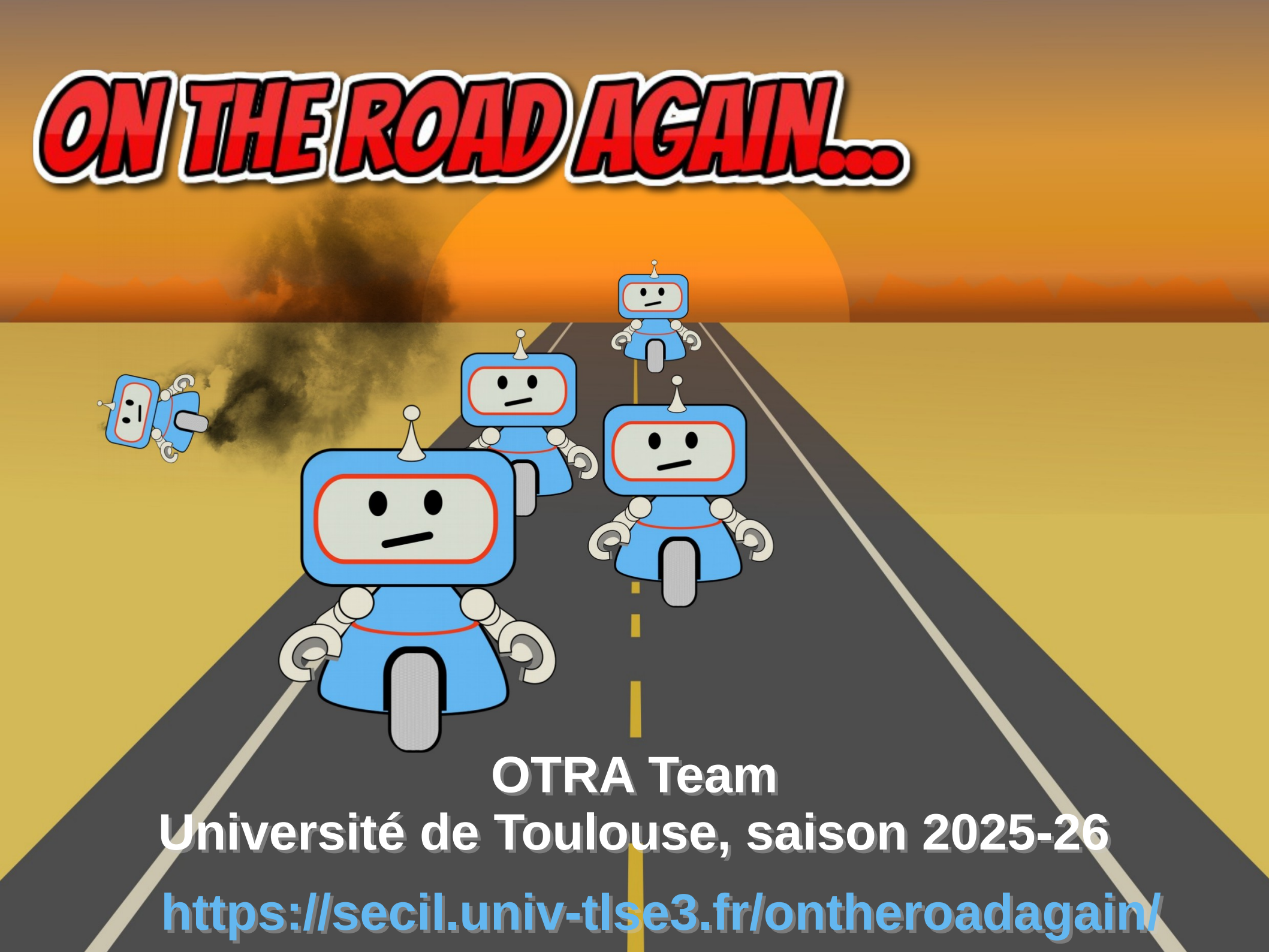
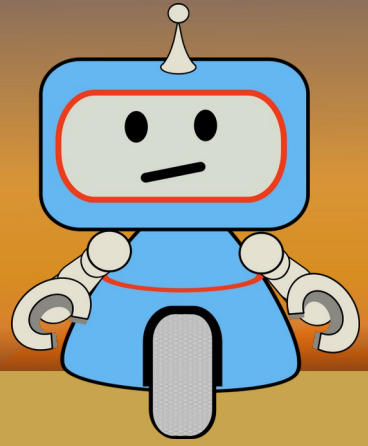
ON THE ROAD AGAIN...



OTRA Team

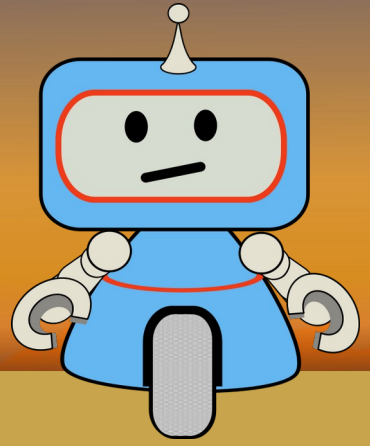
Université de Toulouse, saison 2025-26

https://secil.univ-tlse3.fr/ontheroadagain/



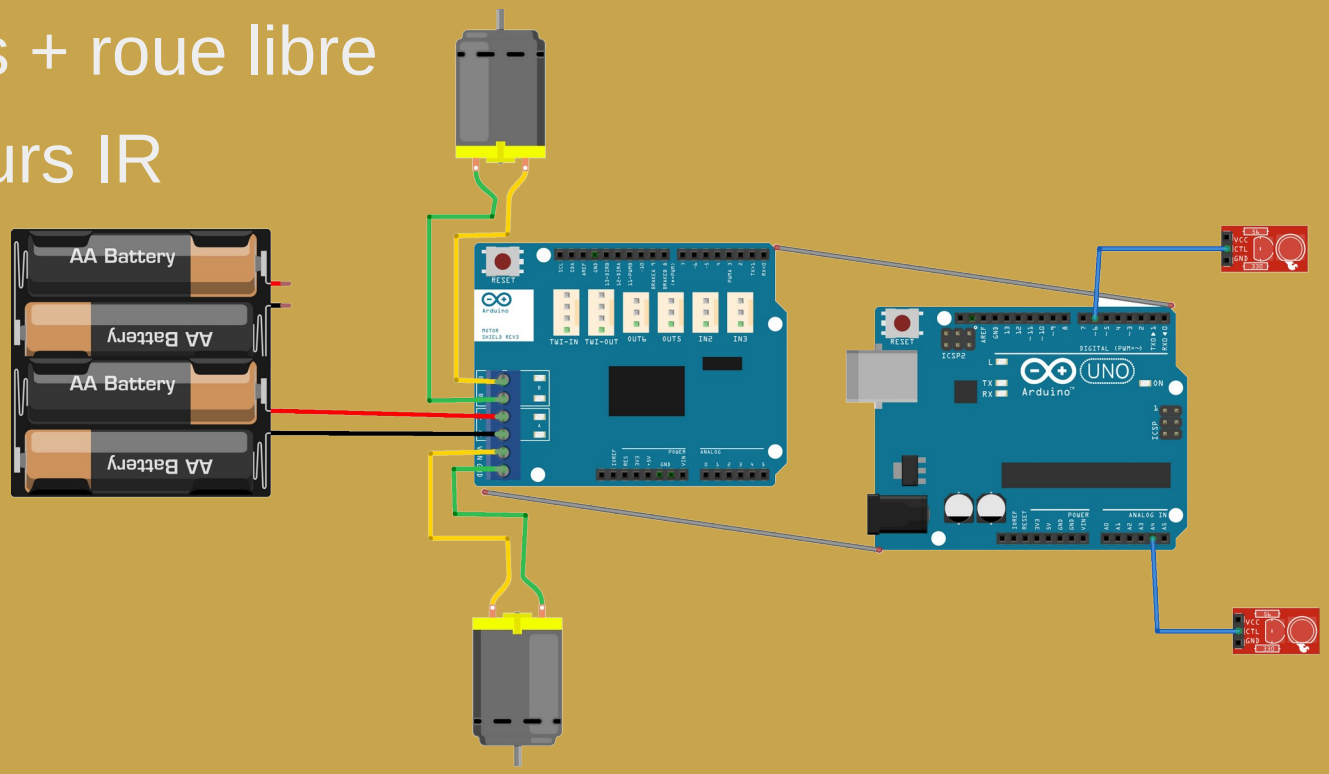
PRINCIPE DU CHALLENGE

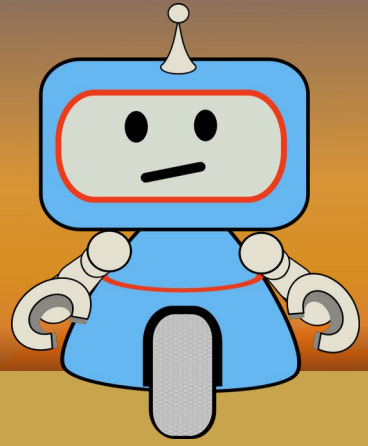
- équipe de 2-3 personne
- réalisation d’un robot suiveur de ligne de A à Z
  - châssis, câblage programmation, test + mise au point
- formation
  - documentation sur le site
  - échanges
- Épreuves du challenge
  - course vitesse
  - course de robustesse



MATERIEL UTILISE

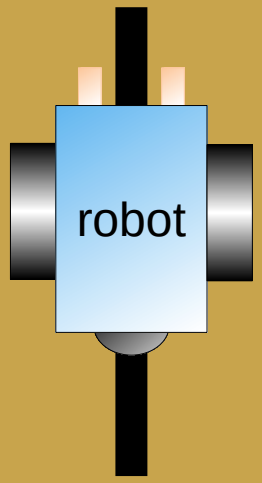
- Matériel fourni
  - calculateur : Arduino + contrôleur moteur
  - 2 moteurs électriques classiques
  - 2 roues motrices + roue libre
  - rampe de capteurs IR
  - kit Arduino
  - alimentation
- Atelier :
  U3-305



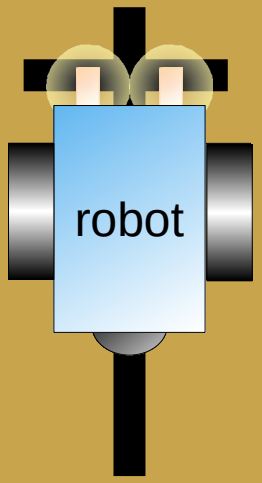


LES CAPTEURS

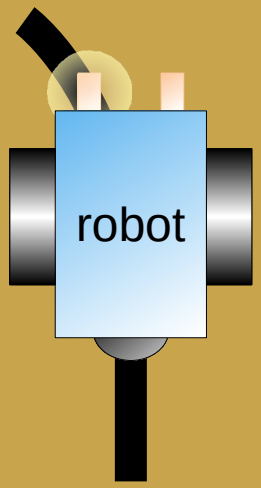
Ligne droite :
gauche → blanc
droit → blanc



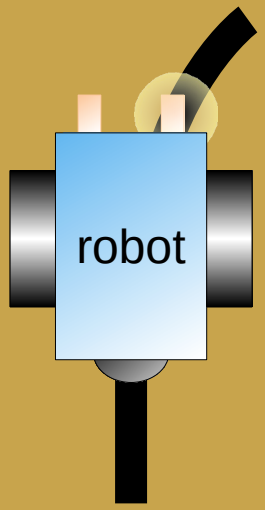
Barre d'arrêt :
gauche → noir
droit → noir

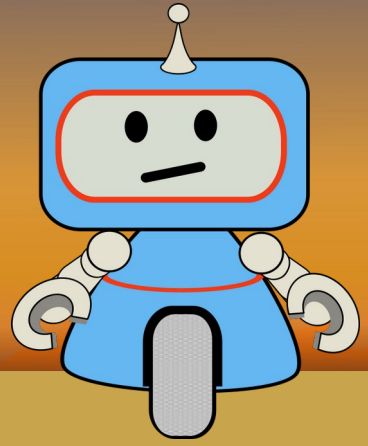


Virage à gauche :
gauche → noir
droit → blanc

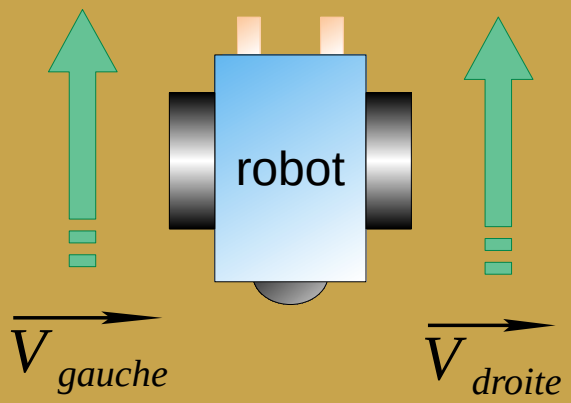


Virage à droite:
gauche → blanc
droit → noir

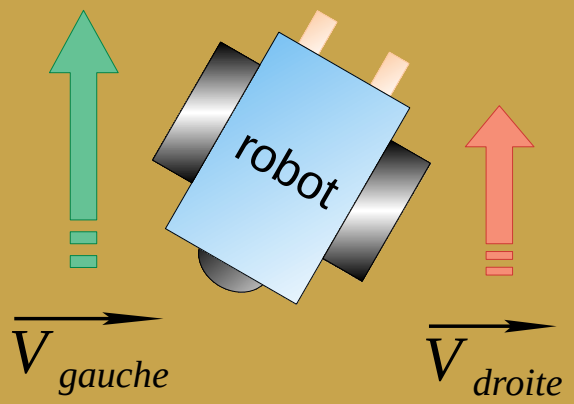




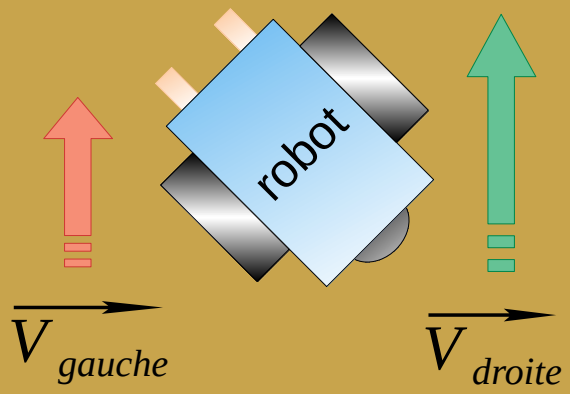
LES MOTEURS



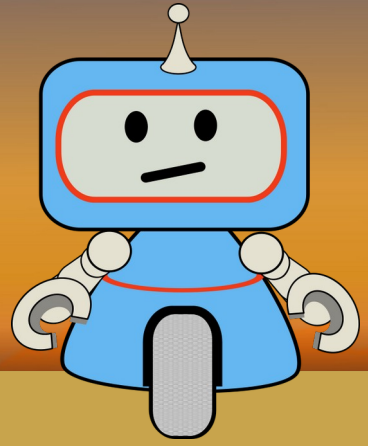
$$\overrightarrow{V}_{gauche} = \overrightarrow{V}_{droite}$$



$$\overrightarrow{V}_{gauche} > \overrightarrow{V}_{droite}$$

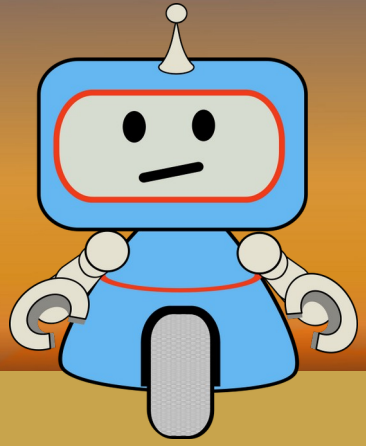


$$\overrightarrow{V}_{gauche} < \overrightarrow{V}_{droite}$$



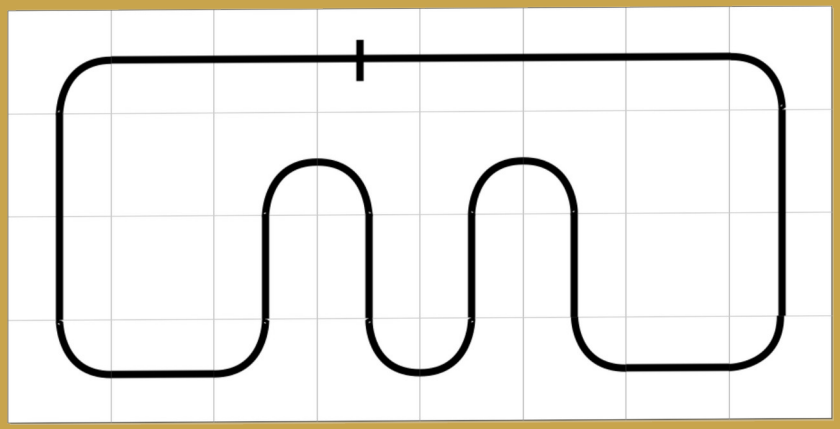
LES EPREUVES

- Épreuve de vitesse
  - trajet simple (quelques tournants)
  - chronométré
- Épreuve de robustesse
  - nombreux tournants, compliqués, intersection
  - + grande distance, + court temps
- Prix
  - plaisir d’avoir réalisé son robot
  - avoir fait son BE
  - Diplôme, T-shirt ? trophée ?

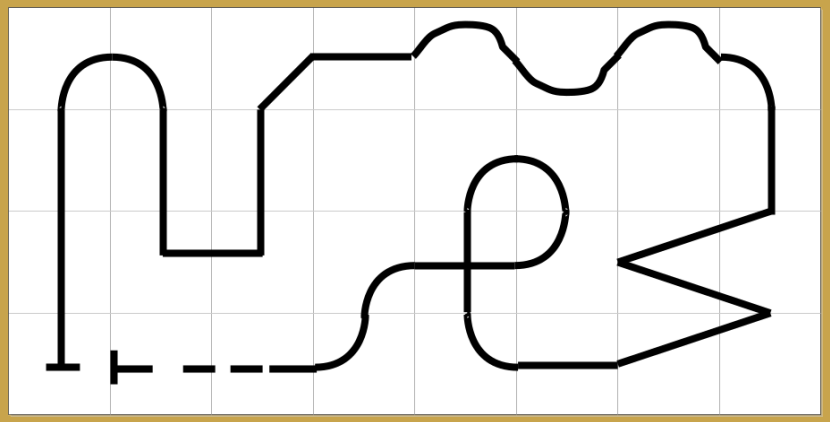


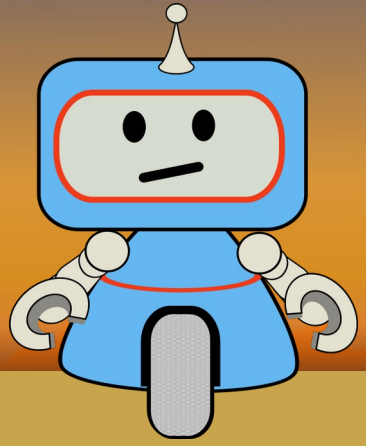
CIRCUITS

Rapidité



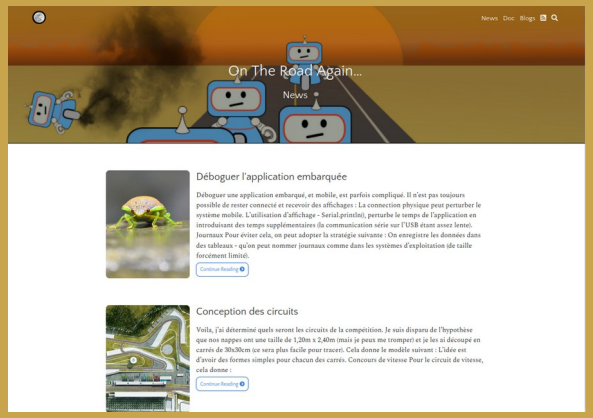
Robustesse



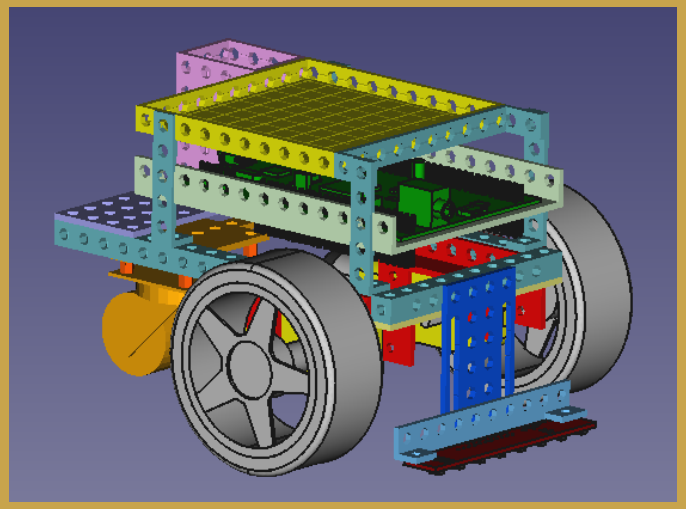


ORGANISATION

- Source d’information : https://secil.univ-tlse3.fr/ontheroadagain/
  - article de documentation
  - Blog, story / groupe (Git + Hugo)
- Local : U3-305
- Réunion d’avancement régulière
- Concours à la fin
- Remise des sources et du matériel



CONSTRUCTION DU ROBOT



- Impression 3D
  - CampusFab
  - Moi-même (Discord)
- Logiciels
  - nombreux logiciels de modélisation
  - OpenScad / FreeCAD
  - bibliothèque RobotParts https://github.com/hcasse/ontheroadagain
  -

D'AUTRES QUESTIONS?

https://secil.univ-tlse3.fr/ontheroadagain/

UNIVERSITE TOULOUSE III PAUL SABATIER
Université de Toulouse
Faculté Sciences et Ingénierie

SIAME + iLoRD = SECIL

D[i] Département Informatique